FastAPI REST API Specification (OpenAPI 3.1)

Waypoint Travel Intelligence Platform · 10 Production Endpoints

Endpoint 1: POST /api/v1/plan-trip

Executes the complete 8-agent LangGraph workflow with input guardrails, flight recommendations, and iterative quality validation.

Sample Request:

```
{
  "query": "5 days trip to Goa with friends, nightlife and beaches for INR 40000",
  "destination_override": null,
  "budget_override": 40000.0,
  "duration_override": 5
}
```

Sample Response (Truncated):

```
{
  "primary_destination": "Goa",
  "budget_analysis": {"estimated_total": 35600, "remaining_budget": 4400, "feasible": true},
  "flights": {"flight_options": [{"airline": "IndiGo", "price_estimate": 4200, "route": "DEL -> GOI"}]},
  "itinerary": {"daily_itinerary": [...]},
  "validation": {"quality_score": 0.93, "approved": true},
  "agent_traces": [...]
}
```

Endpoint 2: POST /api/v1/search

Direct Hybrid Retrieval combining BM25 keyword matching and ChromaDB dense embeddings with metadata filtering.

Endpoint 3: POST /api/v1/recommend

Extracts traveler preferences and ranks candidate destinations with 5-factor score breakdowns.

Endpoint 4: POST /api/v1/itinerary

Direct multi-day chronological itinerary builder given explicit destination, duration, budget, and interest parameters.

Endpoint 5: GET /api/v1/explore

Returns catalog of destinations with ratings computed from historical UCI Dataset 485 traveler scores across themes.

Endpoint 6: POST /api/v1/explore/recommend

Ranks catalog destinations by matching user preference tags against similar historical traveler ratings.

Endpoint 7: POST /api/v1/trip-options

Lists realistic transit options (flight, train, bus) and hotel tiers for a destination from the traveler's origin city.

Endpoint 8: POST /api/v1/build-trip

Generates an end-to-end custom day plan and total price from the traveler's explicit route, lodging, and party choices.

Endpoint 9: POST /api/v1/chat

Conversational AI travel assistant for fast Q&A; on seasons, flights, packing, and destination advice.

Endpoint 10: GET /health

System health check returning database status, 5,456 indexed vectors count, and active LLM configuration.